Notes for 'The Quotient Rule'

Important Ideas and Useful Facts:

(i) The Quotient Rule: Suppose that $y = \frac{u}{v}$ where u and v are differentiable functions of x. Then, using Leibniz notation,

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2},$$

or, equivalently, using function notation,

$$y' = \frac{vu' - uv'}{v^2}.$$

(ii) The derivative of the tangent function: Define the *secant function*

$$\sec x = \frac{1}{\cos x},$$

for all x such that $\cos x$ is nonzero, and write $\sec^2 x$ for $(\sec x)^2$. Then

$$\frac{d}{dx}\big(\tan x\big) = \frac{1}{\cos^2 x} = \sec^2 x.$$

Examples and derivations:

1. We use the Product and Chain Rules to derive the Quotient Rule

$$y' = \frac{vu' - uv'}{v^2}$$

assuming that $y = \frac{u}{v}$.

Proof: We may rewrite y as the product $y = uv^{-1}$, and note that, by the Chain Rule,

$$\big(v^{-1}\big)' = \frac{d}{dx}(v^{-1}) = \frac{d}{dv}(v^{-1})\,\frac{dv}{dx} = -v^{-2}v' = -\frac{v'}{v^2}.$$

By the Product Rule,

$$y' = u(v^{-1})' + v^{-1}u' = u\left(-\frac{v'}{v^2}\right) + \frac{u'v}{v^2} = \frac{vu' - uv'}{v^2},$$

completing the proof.

2. We verify that

$$\frac{d}{dx}\big(\tan x\big) = \frac{1}{\cos^2 x} = \sec^2 x,$$

where $\sec x = \frac{1}{\cos x}$.

Proof: Observe that $\tan x = \frac{u}{v}$ where $u = \sin x$ and $v = \cos x$, so that $u' = \cos x$ and $v' = -\sin x$. By the Quotient Rule,

$$\frac{d}{dx}\big(\tan x\big) = \frac{vu' - uv'}{v^2} = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\sin^2 x + \cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x},$$

using the circular identity in the numerator, which completes the proof.

3. Find y' when $y = \dfrac{x^3+2}{x^2-5}$.

Solution: We have $y = \frac{u}{v}$ where $u = x^3+2$, so that $u' = 3x^2$, and $v = x^2-5$, so that $v' = 2x$. Then, by the Quotient Rule,

$$\begin{aligned} y' &= \frac{vu'-uv'}{v^2} = \frac{(x^2-5)(3x^2)-(x^3+2)(2x)}{(x^2-5)^2} = \frac{3x^4-15x^2-2x^4-4x}{(x^2-5)^2} \\ &= \frac{x^4-15x^2-4x}{(x^2-5)^2} = \frac{x(x^3-15x-4)}{(x^2-5)^2}\,. \end{aligned}$$

4. Find y' when $y = \dfrac{1+e^{-x}}{2+e^{2x}}$.

Solution: We have $y = \frac{u}{v}$ where $u = 1+e^{-x}$, so that $u' = -e^{-x}$, and $v = 2+e^{2x}$, so that $v' = 2e^{2x}$. Then, by the Quotient Rule,

$$\begin{aligned} y' &= \frac{vu'-uv'}{v^2} = \frac{(2+e^{2x})(-e^{-x})-(1+e^{-x})(2e^{2x})}{(2+e^{2x})^2} = \frac{-2e^{-x}-e^{x}-2e^{2x}-2e^{x}}{(2+e^{2x})^2} \\ &= \frac{-2e^{-x}-3e^{x}-2e^{2x}}{(2+e^{2x})^2} = \frac{-e^{-x}(2+3e^{2x}+2e^{3x})}{(2+e^{2x})^2}\,. \end{aligned}$$

5. Find $\dfrac{dy}{dt}$ when $y = \dfrac{e^{3t}}{t^2}$.

Solution: We have $y = \frac{u}{v}$ where $u = e^{3t}$ and $v = t^2$, so that $\frac{du}{dt} = 3e^{3t}$ and $\frac{dv}{dt} = 2t$. By the Quotient Rule,

$$\frac{dy}{dt} = \frac{v\frac{du}{dt}-u\frac{dv}{dt}}{v^2} = \frac{(t^2)(3e^{3t})-(e^{3t})(2t)}{t^4} = \frac{e^{3t}(3t-2)}{t^3}\,.$$

6. Find $\dfrac{dv}{du}$ when $v = \dfrac{\ln u}{u^3}$.

Solution: We have $v = \frac{s}{t}$ where $s = \ln u$ and $t = u^3$, so that $\frac{ds}{du} = \frac{1}{u}$ and $\frac{dt}{du} = 3u^2$.

By the Quotient Rule, regardless of the symbols chosen, we "multiply the bottom by the derivative of the top, take away the top times the derivative of the bottom, and put all over the bottom squared", to get

$$\frac{dv}{du} = \frac{t\frac{ds}{du}-s\frac{dt}{du}}{t^2} = \frac{\frac{u^3}{u}-(\ln u)(3u^2)}{u^6} = \frac{1-3\ln u}{u^4}\,.$$

7. Find y' when $y = \dfrac{xe^x}{\cos x}$.

Solution: We have $y = \frac{u}{v}$ where $u = xe^x$, so that $u' = e^x + xe^x$, and $v = \cos x$, so that $v' = -\sin x$. By the Quotient Rule,

$$y' = \frac{vu'-uv'}{v^2} = \frac{(\cos x)(e^x+xe^x)-(xe^x)(-\sin x)}{\cos^2 x} = \frac{e^x(\cos x + x\cos x + x\sin x)}{\cos^2 x}\,.$$

8. Find $\dfrac{du}{d\theta}$ when $u = \dfrac{\theta^2 - 1}{\sqrt{\theta^2+1}}$.

Solution: We have $u = \frac{v}{w}$ where $v = \theta^2 - 1$, so that $\frac{dv}{d\theta} = 2\theta$, and $w = \sqrt{\theta^2+1} = (\theta^2+1)^{1/2}$, so that, by the Chain Rule,

$$\frac{dw}{d\theta} \;=\; \frac{1}{2}\big(\theta^2+1\big)^{-1/2}(2\theta) \;=\; \frac{\theta}{\sqrt{\theta^2+1}} \; .$$

By the Quotient Rule, regardless of the symbols chosen, we "multiply the bottom by the derivative of the top, take away the top times the derivative of the bottom, and put all over the bottom squared", to get

$$\begin{aligned}
\frac{du}{d\theta} \;&=\; \frac{w\frac{dv}{d\theta} - v\frac{dw}{d\theta}}{w^2} \;=\; \frac{\big(\sqrt{\theta^2+1}\big)(2\theta) - (\theta^2-1)\big(\frac{\theta}{\sqrt{\theta^2+1}}\big)}{\theta^2+1} \\
&=\; \frac{(\theta^2+1)(2\theta) - (\theta^2-1)(\theta)}{(\theta^2+1)^{3/2}} \;=\; \frac{2\theta^3 + 2\theta - \theta^3 + \theta}{(\theta^2+1)^{3/2}} \\
&=\; \frac{\theta(\theta^2+3)}{(\theta^2+1)^{3/2}} \; .
\end{aligned}$$